Vehicle Rental Management System

Object-Oriented Programming Assignment
Prepared By: Ayat

Introduction

The Vehicle Rental Management System is a console-based Java application developed using Object-Oriented Programming principles. The system manages vehicles, customers, rental transactions, maintenance operations, and revenue calculation.

Abstraction

Vehicle was designed as an abstract class to define common properties and behaviors shared by all vehicle types while preventing direct object creation.

Encapsulation

Important attributes were declared private and accessed through getter and setter methods to protect data integrity.

Inheritance

Car, Motorcycle, and Truck inherit from Vehicle to reuse common functionality and reduce code duplication.

Abstract Class vs Interface

Vehicle is an abstract class because all vehicles share common data. Maintainable is an interface that defines maintenance-related behavior.

Polymorphism

Different vehicle types are stored and processed through Vehicle references and the Maintainable interface.

Method Overloading

The project overloads searchVehicle using different parameter lists.

Method Overriding

Car, Motorcycle, and Truck override calculateRentalCost() and displayInfo() methods.

Collections and Data Management

The system uses ArrayList collections to store vehicles, customers, and rentals dynamically.

System Features

The system supports vehicle management, customer registration, rentals, returns, maintenance operations, search functions, and revenue calculation.

Conclusion

The project successfully demonstrates abstraction, encapsulation, inheritance, polymorphism, interfaces, overloading, and overriding using Java OOP principles.